09-26 第1篇

阅读理解（说明文） How Do Bees Turn Flowers into Honey?

① Open a jar of **golden honey** and you open one of nature's most patient stories. On your table sit only a few spoonfuls of thick, sweet liquid; yet each of those spoonfuls has travelled through **thousands of flowers** and the bodies of **many small bees**. How exactly does a bee turn a thin, watery drop of flower juice into the rich honey we spread on our bread? The story has **four careful steps**, and it shows one of the cleverest teamworks in the natural world.

② **Step 1 — The flight.** A worker bee leaves the hive in the early morning and may visit **up to 100 flowers** during a single trip. Each flower offers a small amount of **nectar**, a sweet, watery liquid made by the plant to attract insects. The bee uses its long tongue to suck up the nectar and stores it, not in the main stomach, but in a special second stomach called the **"honey stomach"** or **crop**. The crop can hold almost as much as the bee itself weighs. Bees may fly out of the hive 10 to 15 times a day. To make **just one kilogram** of honey, a whole hive must together visit around **two million flowers** and fly a distance roughly equal to **three times around the Earth**.

③ **Step 2 — The passing.** When the worker bee returns to the hive, she does not put the nectar directly into the wax cells. Instead, she gives it to a **"receiver bee"**, mouth to mouth. The receiver bee chews the nectar for a while and then passes it to another bee, who does the same. Each time nectar passes from one bee to another, special **enzymes** from the bees' own bodies are mixed into it. These enzymes slowly break the big sugars in nectar into **simpler sugars** that will keep for years without going bad.

④ **Step 3 — The drying.** Fresh nectar is still about **70 % water**. If it were stored like that, it would soon **ferment** (turn sour) in the warm hive. So bees must **dry** the nectar. They place small drops of it into the little **wax cells** they have built inside the hive, and then, in a habit that seems almost strange, they **beat their wings rapidly** near the cells, creating a warm, moving breeze. This moving air slowly carries the water away. Over a few days, the nectar becomes thicker and thicker, until its water falls to only about **17 — 18 %** — the level where honey will not ferment and can store for years.

⑤ **Step 4 — The sealing.** When a wax cell is full of thick, finished honey, the bees cover the top with a thin **beeswax cap**, sealing the honey inside. A full hive in a good summer may store **20 to 30 kilograms** of honey — much more than the colony really needs for winter. Many beekeepers, therefore, take out the **extra honey** without harming the colony, leaving enough to feed the bees through cold months.

⑥ A few interesting facts help us understand how special honey really is. First, honey is one of the **few foods in the world that never spoils**; archaeologists have opened ancient Egyptian tombs more than 3 000 years old and found honey still safe to eat. Second, the **colour and flavour** of honey depend on the flowers visited: honey from lavender tastes and looks different from honey made near an orange tree. Third, bees do not only make honey — by flying from flower to flower, they **pollinate** many of the plants that feed us, including apples, cucumbers, almonds and strawberries. Without bees, our dinner tables would be much emptier than most people realise.

⑦ So the next time you open a small jar of honey, **pause for a second**. Inside that soft golden weight lie the careful flights of many bees, the patient mouth–to–mouth chewing of the hive, the cool afternoon wings fanning above the wax cells, and the final quiet cap of beeswax. A spoon of honey is, in a very real way, a **map of two million flowers** — and a reminder that the smallest workers in nature are often the most important.

注释： hive 蜂巢；nectar 花蜜；crop 嗉囊；enzyme 酶；ferment 发酵；beeswax 蜂蜡；pollinate 授粉

根据文章内容，选择最佳答案。

1. About how far must a hive's bees together fly to make just ONE kilogram of honey?

A. About one kilometre in total.
B. Roughly the distance from their city to the next village.
C. About three times around the Earth, visiting around two million flowers.
D. Only the distance between two trees.

2. What happens in Step 2, "The passing" (Paragraph 3)?

A. The worker bee sleeps for a while.
B. The nectar is passed mouth to mouth between bees, and special enzymes from the bees' bodies mix with it, slowly breaking big sugars into simpler sugars.
C. The bees fry the nectar over a small fire.
D. The nectar is thrown out of the hive.

3. Why do bees beat their wings near the wax cells in Step 3 (Paragraph 4)?

A. To scare other insects away.
B. To dance as part of a ceremony.
C. To create a warm, moving breeze that dries the nectar, lowering its water from about 70 % to 17—18 %, so the honey will not ferment.
D. To cool the hive down in summer only.

4. According to Paragraph 6, which is TRUE about honey?

A. Honey spoils within a week and must be eaten quickly.
B. The colour and flavour of honey depend on the kind of flowers the bees visit, and honey is one of the few foods that never spoils — honey found in ancient Egyptian tombs is still safe to eat.
C. Every jar of honey tastes exactly the same.
D. Bees make honey but do not pollinate any plants.

5. What is the main message of the last paragraph?

A. Honey is cheap and not worth thinking about.
B. A spoon of honey is a "map of two million flowers", the result of the careful flights and teamwork of many bees; it reminds us that the smallest workers in nature are often the most important.
C. Students should eat as much honey as possible.
D. Bees have no real effect on human food.

答案与解析

1.
【答案】C
【题型】细节理解题
【定位】第②段 "visit around two million flowers and fly a distance roughly equal to three times around the Earth"
【解析】1 kg 蜂蜜需飞绕地球三圈。A、B、D 均错。故选 C。

2.
【答案】B
【题型】细节理解题
【定位】第③段 "mouth to mouth ... enzymes ... break the big sugars into simpler sugars"
【解析】蜜蜂间口传并加入酶。A、C、D 均错。故选 B。

3.
【答案】C
【题型】细节理解题
【定位】第④段 "beat their wings rapidly ... water ... falls to only about 17—18 %"
【解析】扇翅是为了蒸发水分。A、B、D 均错。故选 C。

4.
【答案】B
【题型】细节理解题
【定位】第⑥段 "colour and flavour ... depend on the flowers ... ancient Egyptian tombs ... still safe to eat"
【解析】蜂蜜几乎不会变质且风味因花而异。A、C、D 均错。故选 B。

5.
【答案】B
【题型】主旨/观点题
【定位】第⑦段 "map of two million flowers ... smallest workers in nature are often the most important"
【解析】末段升华：蜂蜜体现自然小工人的伟大。A、C、D 均偏离。故选 B。